Achintya Rai

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Education

University of California, San Diego — Sept. 2024 – June 2028
Bachelor of Science in Computer Engineering — ***GPA: 3.9/4.0***

Work Experience

Adobe — San Jose, CA
Software Engineering Intern — *June 2026 – Sept. 2026*

- Profiled **60.6M+** payment events in Databricks, flagging **86,387** duplicate records and a **47%** timestamp null rate to establish the authoritative data source for downstream monitoring.
- Designed a time-series anomaly detector spanning **10+** workflows and **30+** countries.
- Developed SQL pipelines and an MLOps workflow to schedule weekly retraining and inference.

VALT Health — Remote
Founding Software Engineer — *Aug. 2025 – Feb. 2026*

- Engineered a Flask backend processing **100k+** datapoints from 20+ fitness-device sources.
- Implemented async batch processing in GCP, raising execution speed **35%** and cutting writes **60%**.
- Automated local setup and data seeding via Bash scripts, reducing testing time up to **80%**.

Decklar (formerly Roambee) — Sunnyvale, CA
Software Engineering Intern — *June 2024 – Aug. 2024*

- Optimized Python and NumPy processing pipelines for real-time shipment monitoring, achieving a **15%** capacity boost across **1M+** weekly events for Amazon, Coca-Cola, and Samsung.
- Implemented advanced data structures for analytics modules, yielding **20%** faster reporting.
- Enhanced workflow efficiency **15%** via Panels-based UI/UX refinements and iterative user testing.

Research

Graph Theory & Neural Architecture — San Diego, CA
Undergraduate Research Fellow (NSF REU) — *March 2026 – Present*

- Engineered a Python pipeline to analyze LLM embedding manifolds, building weight-matrix extraction utilities to compress **500GB** model checkpoints into **13GB** localized manifolds.
- Accelerated preprocessing scripts **30x** (30 min to <1 min), enabling analysis on **8GB RAM** systems.
- Applied Rent's Rule and Louvain clustering to raise ICN density **0.18%** to **13.25%**.

Projects

SquaredUpChess | *React, TypeScript, Django, WebSockets, AWS, Docker* — Aug. 2025 – Dec. 2025

- Engineered a full-stack platform with **React**/**TypeScript** and a **Django REST** backend for real-time, low-latency lesson tracking, scheduling, and payment processing via **WebSockets**.
- Established a **CI/CD pipeline** with automated testing to ensure **99.9%** deployment reliability.
- Optimized **SQLite** via indexing and query tuning to sustain multi-user scalability under load.

Watchdog | *Python, Gemini API, Browser-Use, Asyncio, SQLite* — April 2026

- Engineered an agentic browser safeguard using **browser-use** and **Gemini Vision** to monitor tabs.
- Developed a concurrent **Asyncio** CLI for real-time URL blacklisting and **Notion API** task sync.

NanoGPT | *Python, PyTorch, Transformers, Neural Networks* — Mar. 2026

- Implemented a transformer from scratch in **PyTorch** with multi-headed attention and backprop.
- Engineered scalable dataloaders to stream tokenized data, sustaining **100%** hardware utilization.

Technical Skills

Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL (Postgres/SQLite), Bash, Data Structures & Algorithms
Development: React, Django, Node.js, FastAPI, Flask, Asyncio, WebSockets, REST APIs, CI/CD, PyTest
AI & Cloud: PyTorch, TensorFlow, AWS, GCP, Gemini/OpenAI API, MCP, Docker, GitHub Actions, Linux